Groups and Lie Algebras: Exercise Sheet 3

Exercise 3.1: Some properties of characters

Let V and W be a finite dimensional representation of a finite group G with characters χ_V and χ_W. Compute the character of the representations

$$V \oplus W, \quad V \otimes W, \quad V^*, \quad \mathrm{Hom}(V,W), \quad \Lambda^2 V, \quad \mathrm{Sym}^2 V.$$

Answer:

To compute the characters, we use the definitions of the above representations and the properties of the trace:

- The direct sum $V \oplus W$ is defined as the vector space $V \oplus W = \{(v,w) \mid v \in V, w \in W\}$ with the action of G given by $g \cdot (v,w) = (g \cdot v, g \cdot w)$. The trace of this representation is the sum of the traces of the individual representations, so we have $\chi_{V \oplus W}(g) = \chi_V(g) + \chi_W(g)$.
- The tensor product $V \otimes W$ is defined as the vector space generated by the elements $v \otimes w$ with $v \in V$ and $w \in W$, with the action of G given by $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$. The trace of this representation is the product of the traces of the individual representations, so we have $\chi_{V \otimes W}(g) = \chi_V(g) \cdot \chi_W(g)$.
- The dual representation V^* is defined as the vector space of linear functionals on V, with the action of G given by $(g \cdot f)(v) = f(g^{-1} \cdot v)$ for $f \in V^*$ and $v \in V$. The trace of this representation is the complex conjugate of the trace of the original representation, so we have $\chi_{V^*}(g) = \overline{\chi_V(g)}$.
- The verctor space of homomorphisms $\mathrm{Hom}(V,W)$ is the set of maps from V to W that preserve the group structure. The action of G is given by $(g \cdot \phi)(v) = g_W \cdot \phi(g_V^{-1} \cdot v)$. To compute the character we first need to show $\mathrm{Hom}(V,W) \cong V^* \otimes W$. To show this we define the map

  $$\Phi : V^* \otimes W \to \mathrm{Hom}(V,W), \quad (\phi \otimes w) \mapsto (v \mapsto \phi[v] \cdot w)$$

  This is indeed an isomorphism since

  $$\dim(V^* \otimes W) = \dim(V) \cdot \dim(W) = \dim(\mathrm{Hom}(V,W))$$

To show that Φ is injective, we evaluate the zero map on the basis vectors e_k of V:

$$\begin{aligned}\Phi\left(\sum_{i,j} a_{ij}\,\phi_i \otimes w_j\right)[e_k] &= 0\\ \sum_{i,j} a_{ij}\,\phi_i[e_k]\cdot w_j &= 0\\ \sum_{i,j} a_{ij}\,\delta_{ik}\cdot w_j &= 0\\ \sum_{j} a_{kj}\,w_j &= 0\end{aligned}$$

Since the w_j are linearly independent, we have $a_{kj} = 0$ for all j. Since this is true for all k, we have $a_{ij} = 0$ for all i, j and thus Φ is injective. Therefore we have $\mathrm{Hom}(V, W) \cong V^* \otimes W$ and we can compute the character as $\chi_{\mathrm{Hom}(V,W)}(g) = \chi_{V^*\otimes W}(g) = \chi_{V^*}(g)\cdot\chi_W(g) = \overline{\chi_V(g)}\cdot\chi_W(g)$.

- To compute the characters of the symmetric and antisymmetric rank 2 tensors, we notice that

  $$V \otimes V = \Lambda^2 V \oplus \mathrm{Sym}^2 V$$

  This can be seen by simply writing down the definitions. For all $v, w \in V$ we have

  $$v \otimes w = \frac{1}{2}(v \otimes w + w \otimes v) + \frac{1}{2}(v \otimes w - w \otimes v) = v \vee w + v \wedge w$$

  Therefore we can compute the characters of the symmetric and antisymmetric rank 2 tensors as

  $$\chi_{V\otimes V}(g) = \chi_{\Lambda^2 V}(g) + \chi_{\mathrm{Sym}^2 V}(g)$$

  Futhermore consicer the eigenvalues of the representation V for a fixed $g \in G$. Let these eigenvalues be $\lambda_1, \ldots, \lambda_n$. Then we have

  $$\chi_{V\otimes V}(g) = \sum_{i,j}\lambda_i\lambda_j$$

  Futhermore, if λ_i are the eigenvalues of g in V, then the eigenvalues of g^2 in V are λ_i^2. Therefore we have

  $$\chi_V(g^2) = \sum_i \lambda_i^2$$

  We can now compute the characters of the symmetric and antisymmetric rank 2 tensors as

  $$\chi_{\Lambda^2 V}(g) = \sum_{i<j}\lambda_i\lambda_j = \frac{1}{2}\left(\sum_{i,j}\lambda_i\lambda_j - \sum_i \lambda_i^2\right) = \frac{1}{2}(\chi_{V\otimes V}(g) - \chi_V(g^2))$$

  $$\chi_{\mathrm{Sym}^2 V}(g) = \sum_{i\leq j}\lambda_i\lambda_j = \frac{1}{2}\left(\sum_{i,j}\lambda_i\lambda_j + \sum_i \lambda_i^2\right) = \frac{1}{2}(\chi_{V\otimes V}(g) + \chi_V(g^2))$$

$\blacksquare$

Exercise 3.2: Quaternions

The quaternion group consists of 8 elements

$$\pm 1, \quad \pm i, \quad \pm j, \quad \pm k.$$

The defining relations are

$$i^2 = j^2 = k^2 = -1, \qquad i = jk = -kj, \qquad j = ki = -ik, \qquad k = ij = -ji.$$

a) First find 1 four-dimensional representations of Q_8.

Answer:

We start by writing the elements of Q_8 as the vectors

$$1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad i = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad j = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad k = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

We can then define a representation by left multiplication. To find the explicit form we use the above relations. 1 acts as the identity, so we start by considering the action of i:

$$\rho(i) \cdot 1 = 1, \quad \rho(i) \cdot i = 1, \quad \rho(i) \cdot j = k, \quad \rho(i) \cdot k = -j$$

so we have

$$\rho(i) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

We can do the same for j and k:

$$\rho(j) = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad \rho(k) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

To check if this representation is irreducible, we compute the inner product of the character with itself:

$$\langle \chi, \chi \rangle = \frac{1}{|Q_8|} \sum_{g \in Q_8} |\chi(g)|^2 = \frac{1}{8}(2 \cdot 4^2 + 6 \cdot 0^2) \neq 1$$

This means this representation is not irreducible. ∎

b) Find 4 one-dimensinal representations of Q_8. Are they irreducible?

Answer:

To find the one-dimensional representations, we find a homomorphism from Q_8 to $\mathbb{C}^\times$. We start by considering $\rho(1) = 1$ and $\rho(-1)^2 = \rho(1) = 1$. This means $\rho(-1) \in \{\pm 1\}$. Similarly, we have $\rho(i)^2 = \rho(j)^2 = \rho(k)^2 = \rho(-1)$. Thus if $\rho(-1) = 1$ then we have $\rho(i) = \rho(j) = \rho(k) \in \{\pm 1\}$ and if $\rho(-1) = -1$ then we have $\rho(i), \rho(j), \rho(k) \in \{\pm i\}$. From the relations above we can exclude $\rho(-1) = -1$ since $\rho(i) = \rho(j)\rho(k) = \pm i^2 \notin \{\pm i\}$. Thus we can write down all one-dimensional representations by choosing $\rho(i), \rho(j), \rho(k) \in \{\pm 1\}$. But since $\rho(k) = \rho(i)\rho(j)$, only four choices are possible. We can write them explicitly in the following table:

	1	-1	i	j	k
ρ_1	1	1	1	1	1
ρ_2	1	1	-1	-1	1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	1	-1	-1

They are all irreducible since they are one-dimensional. ■

c) There is also a 2-dimensional irreducible representation. To find it, think of quantum mechanics.

Answer:

To find the 2 dimensional representation, we use the pauli matrices. The pauli-matrices are unitary and self-adjoint, therefore we multiply them by i to get a representation of Q_8:

$$\rho(i) = i\sigma_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \rho(j) = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho(k) = i\sigma_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

Again we check if this representation is irreducible by computing the inner product of the character with itself:

$$\langle \chi, \chi \rangle = \frac{1}{|Q_8|} \sum_{g \in Q_8} |\chi(g)|^2 = \frac{1}{8}(2 \cdot 2^2 + 6 \cdot 0^2) = 1$$

This means this representation is irreducible. ■

d) Are these all irreducible representations? Explain.

Answer:

To check if wa have all irreducible representations, we use the following two facts:

- $|\operatorname{Conj(G)}| = \sum_{\rho \text{ irrep.}} 1$
- $|G| = \sum_{\rho \text{ irrep.}} (\dim \rho)^2$

First we compute the conjugacy classes of Q_8:

$$\operatorname{Conj}(1) = \{g \in Q_8 \mid g = q1q^{-1}\} = \{1\}$$

$$\operatorname{Conj}(-1) = \{g \in Q_8 \mid g = q(-1)q^{-1}\} = \{-1\}$$

Are trivialy given. From the defining relations we can also compute the conjugacy classes of i, j and k:

$$i = jk \quad j = ki \implies -j = -ki = -kjk = kjk^{-1}$$

Similarly we can compute all conjugations of i, j and k and we find that

$$\mathrm{Conj}(i) = \{i, -i\}, \quad \mathrm{Conj}(j) = \{j, -j\}, \quad \mathrm{Conj}(k) = \{k, -k\}$$

So we have 5 conjugacy classes, which means we need 5 irreducible representations. Therefore the 4 one-dimensional representations and the 2-dimensional representation we found are all irreducible representations of Q_8.

We can also derive the fact that the irreducible representations of Q_8 are four 1-dimensional and one 2-dimensional representation using the second fact above. We have

$$|Q_8| = 8 = 2 \cdot 2^2 = 2^2 + 4 \cdot 1^2 = 8 \cdot 1^2$$

Since we need five irreducible representations, we can only have four 1-dimensional and one 2-dimensional representation. ■

e) Compute the character table.

Answer:

We can now compute the character table of Q_8 using the representations we found and the conjugacy classes:

	$\{1\}$	$\{-1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
ρ_1	1	1	1	1	1
ρ_2	1	1	-1	-1	1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	1	-1	-1
ρ_5	2	-2	0	0	0

■

Exercise 3.3: The symmetric group S_4 and its representations

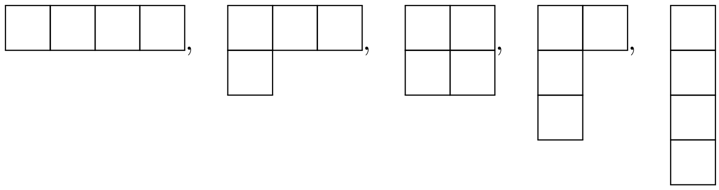
a) Determine the conjugacy classes of S_4 and associate Young diagrams and conjugacy classes.

Answer:

The conjugacy classes of S_4 are given by the cycle types:

- $(id) \Rightarrow (1,1,1,1)$
- $(12) \Rightarrow (2,1,1)$
- $(12)(34) \Rightarrow (2,2)$
- $(123) \Rightarrow (3,1)$
- $(1234) \Rightarrow (4)$

The associated Young diagrams are:



b) There are two 1-dimensional representations of S_4. What are they?

Answer:

A 1-dimensional representation is a homomorphism $\rho : S_4 \to R^*$. Since $\rho(id) = 1$ and $(ij)^2 = id$ where (ij) is a transposition, we have $\rho((ij))^2 = 1$ and thus $\rho((ij)) \in \{\pm 1\}$. Now concider how conjugation acts on a transposition:

$$\tau(ij)\tau^{-1} = (\tau(i)\tau(j))$$

Thus all transpositions are conjugate, therefore all transpositions must be mapped to the same element in $\{\pm 1\}$. Therefore we have two 1-dimensional representations, one where all transpositions are mapped to 1 and one where all transpositions are mapped to -1. The first one is the trivial representation and the second one is the alternating representation.

c) Consider the 4-dimensional standard representation of S_4 on $\mathbb{R}^4$. Decompose it into a 3-dimensional irreducible representation and a 1-dimensional representation.

Answer:

The 4-dimensional standard representation $\rho_{std} : S_4 \to \mathbb{R}^4$ is given by permuting the basis vectors:

$$\rho(\sigma) \triangleright e_i = e_{\sigma(i)}$$

This has a trivial representation as a subrepresentation, given by the vector $v_0 = e_1 + e_2 + e_3 + e_4$. We thus need to find the orthogonal complement $V_\perp$. A choice for this is the vector space:

$$V_\perp = \{x = (x_1, x_2, x_3, x_4)^T \mid x_1 + x_2 + x_3 + x_4 = 0\}$$

This is an invariant subspace since permuting the components does not change the sum of the components. It is also a complement, which can be checked by considering the intersection of $V_\perp$ and the trivial representation:

$$V_\perp \cap \mathrm{span}(v_0) = \{0\}$$

Therefore we have the decomposition $\rho_{std} = \rho_{triv} \oplus \rho_\perp$. To check if $\rho_\perp$ is irreducible, we compute the characters. Since the character of a direct sum is the sum of characters, we have $\chi_{std} = \chi_{triv} + \chi_\perp$. The character of the trivial representations is $\chi_{triv}(\sigma) = 0$ for all σ.The character of the standard representation is given by the number of fixed elements of the permutation:

- $\chi_{std}(id) = 4$
- $\chi_{std}((12)) = 2$
- $\chi_{std}((12)(34)) = 0$
- $\chi_{std}((123)) = 1$
- $\chi_{std}((1234)) = 0$

The inner product of the character of $\rho_\perp$ with itself is given by

$$\langle \chi_\perp, \chi_\perp \rangle = \frac{1}{|S_4|} \sum_{\sigma \in S_4} |\chi_\perp(\sigma)|^2 = \frac{1}{4!}(3^2 + 6 \cdot 1^2 + 3 \cdot (-1)^2) + 8 \cdot 0^2 + 6 \cdot (-1)^2) = 1$$

Thus $\rho_\perp$ is irreducible. ■

d) Take the tensor product of the 3-dimensional representation of the previous part and the alternating representation. It is again irreducible. How many other irreducible representations are there? What are their dimensions?

Answer:

The tensor product is given by:

$$\left(\mathbb{R}^* \otimes \mathbb{R}^3, \, (\rho_{alt} \otimes \rho_\perp)(g) = \rho_{alt}(g) \otimes \rho_\perp(g)\right)$$

The character of a tensor product is the product of the characters, so we have $\chi_{alt \otimes \perp} = \chi_{alt} \cdot \chi_\perp$. The character of the alternating representation is given by the sign of the permutation:

- $\chi_{alt}(id) = 1$
- $\chi_{alt}((12)) = -1$
- $\chi_{alt}((12)(34)) = 1$
- $\chi_{alt}((123)) = 1$
- $\chi_{alt}((1234)) = -1$

Therefore we have:

- $\chi_{alt \otimes \perp}(id) = 3$
- $\chi_{alt \otimes \perp}((12)) = -1$
- $\chi_{alt \otimes \perp}((12)(34)) = -1$
- $\chi_{alt \otimes \perp}((123)) = 0$
- $\chi_{alt \otimes \perp}((1234)) = 1$

Since there are 5 conjugacy classes, there are 5 irreducible representations in total, we have found 4. There is another.

The dimension of the last representation can be computed using the fact that the sum of the squares of the dimensions of the irreducible representations is equal to the order of the group:

$$|S_4| = 4! = 24 = 3^2 + 3^2 + 1^2 + 1^2 + d^2$$

Therefore we have $d = 2$. ■

e) Write down the character table.

Answer:

We can conclude on the characters of the missing representation by writing down the character table and using the fact that the sum of the colu:

	$\{id\}$	$\{(12)\}$	$\{(12)(34)\}$	$\{(123)\}$	$\{(1234)\}$
ρ_{triv}	1	1	1	1	1
ρ_{alt}	1	-1	1	1	-1
$\rho_{\perp}$	3	1	-1	0	-1
$\rho_{\perp\otimes alt}$	3	-1	-1	0	1
ρ_{2d}	2	0	2	-1	0

■

Exercise 3.4: Particle in a box

The Hilbert space of a particle in a box of length l is $L^2([-l,l]^2)$ with the boundary conditions

$$\psi|_{\partial[-l,l]^2} = 0$$

and Hamiltonian

$$\mathbb{H} = -\frac{\hbar^2}{2m}\left(\partial_x^2 + \partial_y^2\right).$$

Show that this system has D_4 symmetry. Solve the Schrödinger equation and find an energy level whose eigenfunctions transform in the 2-dimensional representation of D_4. What is the lowest energy level that does not transform in an irreducible representation of D_4.

Answer:

The symmetry group D_4 contains rotations r and reflections s satisfying the relations $r^4 = s^2 = e$ and $srs = r^{-1}$. We find a representation of D_4 on the Hilbert space by defining the action of r and s on the wavefunction $\psi(x,y)$:

$$(r \triangleright \psi)(x,y) = \psi(-y,x), \quad (s \triangleright \psi)(x,y) = \psi(x,-y)$$

Which satisfies the relations above. To show that this is a symmetry of the system, we show that it commutes with the Hamiltonian. We show the following relations:

$$\partial_x \psi(-y,x) = \left(\frac{\partial r(x)}{\partial x}\partial_x + \frac{\partial r(x)}{\partial y}\partial y\right)\psi(-y,x) = -\partial_y \psi(-y,x)$$

And similarly we can show that $\partial_y \psi(-y,x) = \partial_x \psi(-y,x)$. Therefore we have

$$\mathbb{H}(r \triangleright \psi) = -\frac{\hbar^2}{2m}\left(\partial_x^2 + \partial_y^2\right)\psi(-y,x) = -\frac{\hbar^2}{2m}\left(\partial_y^2 + \partial_x^2\right)\psi(-y,x) = r \triangleright (\mathbb{H}\psi)$$

And therefore $[\mathbb{H}, r] = 0$.And mutatis mutandis we have $[\mathbb{H}, s] = 0$

To solve the Schrödinger equation, we use separation of variables. We write $\psi(x,y) = X(x)Y(y)$ and we find that X and Y satisfy the following equations:

$$-\frac{\hbar^2}{2m}X''(x) = E_x X(x), \quad -\frac{\hbar^2}{2m}Y''(y) = E_y Y(y)$$

With the boundary conditions $X(-l) = X(l) = 0$ and $Y(-l) = Y(l) = 0$. We take an trigonometric ansatz:

$$X(x) = A\sin(k_x x) + B\cos(k_x x), \quad Y(y) = C\sin(k_y y) + D\cos(k_y y)$$

Plugin this into the Schrödinger equation we find that $E_x = \frac{\hbar^2 k_x^2}{2m}$ and $E_y = \frac{\hbar^2 k_y^2}{2m}$. The boundary conditions read as follows:

$$X(-l) = -A\sin(k_x l) + B\cos(k_x l) = 0, \quad X(l) = A\sin(k_x l) + B\cos(k_x l) = 0$$

$$Y(-l) = -C\sin(k_y l) + D\cos(k_y l) = 0, \quad Y(l) = C\sin(k_y l) + D\cos(k_y l) = 0$$

From the first two equations we find that $B = 0$ and $k_x l = n_x \pi$ for some $n_x \in \mathbb{Z}$. From the last two equations we find that $D = 0$ and $k_y l = n_y \pi$ for some $n_y \in \mathbb{Z}$. Therefore we have

$$\psi_{n_x,n_y}(x,y) = A\sin\left(\frac{n_x \pi x}{l}\right)\sin\left(\frac{n_y \pi y}{l}\right)$$

With energy $E_{n_x,n_y} = \frac{\hbar^2\pi^2}{2ml^2}(n_x^2 + n_y^2)$. We find a 2-dimensional representation by considering the vector space spanned by ∎